

LAPORAN SETUP VPS

LAS WEEK 4 (CS)

MBC LAB 2026



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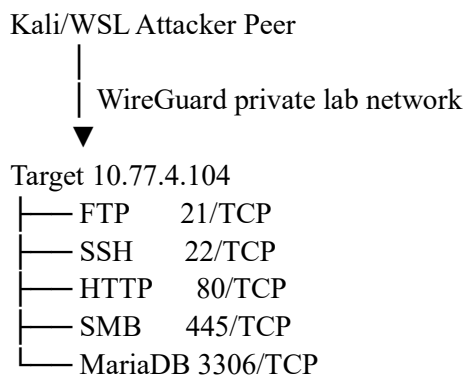
1. Informasi Umum

Item	Detail
Nama Challenge	Midnight Pitlane
Kategori	Linux Privilege Escalation
Platform VPS	Microsoft Azure
Sistem Operasi	Ubuntu Server 24.04 LTS
IP WireGuard	10.77.4.104/32
Interface WireGuard	Mbc-las-w4

2. Tujuan

Lab pengujian ini dibuat agar para kandidat calon asisten MBC lab dapat mempelajari bagaimana cara melakukan service enumeration sampai controlled Linux privilege-escalation proof. Challenge ini di diatur agar para kandidat tidak dapat menggunakan bruteforce, reverse shell, kernel exploit, atau tindakan yang dapat mengubah shared state.

3. Arsitektur



Terdapat public IP Azure yang hanya dapat digunakan sebagai akses administrator pembuat. Karena IP yang di gunakan menggunakan WireGuard private lab network, maka vulnerable service tidak diekspos langsung ke public.

4. Penyediaan VPS Azure

VPS di buat dengan spesifikasi berikut:

- Ubuntu server 24.04 LTS
- 2 vCpu
- 1 GiB memori
- 20 GB disk
- Satu public IPv4

- Network Security Group

5. Verifikasi Jaringan

Perintah berikut di gunakan untuk melihat status jaringan:

```
SupraFast@JustTry:~$ ip -br address
lo                UNKNOWN        127.0.0.1/8 ::1/128
eth0              UP                10.1.1.4/24 metric 100 fe80::7eed:8dff:fef5:e019/64
mbc-las-w4        UNKNOWN        10.77.4.104/32
SupraFast@JustTry:~$ |
```

IP yang diberikan kepada pemain adalah 10.77.4.104, bukan public IP Azure dan bukan private IP Azure 10.1.1.4.

6. Instalasi Paket

```
SupraFast@JustTry:~$ sudo apt update
sudo apt install -y \
  nginx \
  vsftpd \
  samba \
  smbclient \
  mariadb-server \
  mariadb-client \
  gcc \
  libcap2-bin \
  libimage-exiftool-perl \
  curl \
  wget \
  ufw
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1232 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [288 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1689 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5740 B]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Packages [981 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-security/main Translation-en [210 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1205 kB]
```

Paket yang di install adalah paket yang akan di gunakan untuk membuat challenge ini.

7. Pembuatan Akun Challenge

Akun service dibuat terpisah dari akun administrator Azure.

```
SupraFast@JustTry:~$ sudo useradd --create-home --shell /usr/sbin/nologin suprafast
useradd: user 'suprafast' already exists
SupraFast@JustTry:~$ sudo useradd --create-home --shell /usr/sbin/nologin gazzoracing
useradd: user 'gazzoracing' already exists
SupraFast@JustTry:~$ sudo useradd --create-home --shell /bin/bash gtracing
useradd: user 'gtracing' already exists
SupraFast@JustTry:~$ |
```

Untuk password masing masing akun diatur sesuai mentor-only notes. Dan akun gtracing diatur agar tidak berada dalam grup yang sensitif:

```
SupraFast@JustTry:~$ id gtracing
uid=1003(gtracing) gid=1003(gtracing) groups=1003(gtracing)
SupraFast@JustTry:~$ sudo gpasswd -d gtracing sudo 2>/dev/null || true
sudo gpasswd -d gtracing adm 2>/dev/null || true
sudo gpasswd -d gtracing docker 2>/dev/null || true
sudo gpasswd -d gtracing lxd 2>/dev/null || true
sudo gpasswd -d gtracing disk 2>/dev/null || true
sudo gpasswd -d gtracing shadow 2>/dev/null || true
Removing user gtracing from group sudo
Removing user gtracing from group adm
Removing user gtracing from group lxd
Removing user gtracing from group disk
Removing user gtracing from group shadow
SupraFast@JustTry:~$ |
```

8. Konfigurasi HTTP

Website disimpan pada:

/var/www/midnight

Struktur:

/var/www/midnight/

```
|— index.html
|— robots.txt
|— garage/
    |— index.html
    |— supra-inspection.webp
```

Isi robots.txt:

```
SupraFast@JustTry:~$ curl http://10.77.4.104/robots.txt
User-agent: *
Disallow: /garage/
SupraFast@JustTry:~$ |
```

Nginx dikonfigurasi pada /etc/nginx/sites-available/midnight:

```
SupraFast@JustTry:~$ cat /etc/nginx/sites-available/midnight
server {
    listen 10.77.4.104:80;
    server_name _;

    root /var/www/midnight;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }

    autoindex off;
}

SupraFast@JustTry:~$ |
```

Aktivasi:

```
SupraFast@JustTry:~$ sudo rm -f /etc/nginx/sites-enabled/default
sudo ln -sf /etc/nginx/sites-available/midnight /etc/nginx/sites-enabled/midnight
sudo nginx -t
sudo systemctl enable --now nginx
sudo systemctl restart nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
SupraFast@JustTry:~$ |
```

Gambar inspeksi Supra digunakan sebagai clue menuju layanan FTP. Clue disimpan pada metadata XMP/EXIF agar pemain perlu memeriksa informasi file, bukan hanya visualnya.

- Menyiapkan Gambar Supra

Gambar disiapkan menggunakan format webp yang tersedia di dalam halaman HTTP. Kemudian setelah disiapkan gambar diunggah ke VPS melalui SCP.

```
Ilyas@HyperX mnt/ .. ➡ ../ilyas >$ scp supra-inspection.webp \
SupraFast@10.77.4.104:/tmp/supra-inspection.webp
```

Di VPS, salin gambar ke direktori galeri website:

```
SupraFast@JustTry:~$ sudo cp /tmp/supra-inspection.webp \
/var/www/midnight/garage/supra-inspection.webp
```

Pastikan file tersedia di VPS:

```
SupraFast@JustTry:~$ file /var/www/midnight/garage/supra-inspection.webp
/var/www/midnight/garage/supra-inspection.webp: RIFF (Little-endian) data, Web/P image, VP8 encoding, 900x600, Scaling:
[none]x[none], YUV color, decoders should clamp
SupraFast@JustTry:~$ ls -lh /var/www/midnight/garage/supra-inspection.webp
-rw-r--r-- 1 root root 59K Sep  2 01:22 /var/www/midnight/garage/supra-inspection.webp
SupraFast@JustTry:~$ |
```

- Menambahkan Metadata dengan ExifTool

ExifTool dipasang untuk menambahkan metadata ke dalam gambar. Kemudian tambahkan deskripsi gambar dan petunjuk FTP menggunakan perintah berikut:

```
SupraFast@JustTry:~$ sudo exiftool \
-overwrite_original \
-XMP-dc:Creator='SupraFast' \
-XMP-dc:Title='MK4 Pre-Race Inspection' \
-XMP-dc:Description='Archive service: FTP | Crew: suprafast | Access phrase: ToyotaMK42JZ' \
-XMP:Instructions='Use the legacy FTP service to access the pit archive.' \
/var/www/midnight/garage/supra-inspection.webp
```

Opsi -overwrite_original mencegah ExifTool menyimpan file cadangan berekstensi _original. Setelah proses selesai, verifikasi seluruh metadata:

```
SupraFast@JustTry:~$ exiftool -s -G1 -s \
/var/www/midnight/garage/supra-inspection.webp
[ExifTool] ExifToolVersion : 12.76
[System] FileName : supra-inspection.webp
[System] Directory : /var/www/midnight/garage
[System] FileSize : 63 kB
[System] FileModifyDate : 2026:09:02 01:29:01+00:00
[System] FileAccessDate : 2026:09:02 01:29:01+00:00
[System] FileInodeChangeDate : 2026:09:02 01:29:01+00:00
[System] FilePermissions : -rw-r--r--
[File] FileType : Extended WEBP
[File] FileTypeExtension : webp
[File] MIMEType : image/webp
[RIFF] WebP_Flags : XMP
[RIFF] ImageWidth : 900
[RIFF] ImageHeight : 600
[RIFF] VP8Version : 0 (bicubic reconstruction, normal loop)
[RIFF] ImageWidth : 900
[RIFF] HorizontalScale : 0
[RIFF] ImageWeight : 600
[RIFF] VerticalScale : 0
[XMP-x] XMPToolkit : Image::ExifTool 12.76
[XMP-dc] Creator : SupraFast
[XMP-dc] Description : Archive service: FTP | Crew: suprafast | Access phrase: ToyotaMK42JZ
[XMP-dc] Title : MK4 Pre-Race Inspection
[XMP-photoshop] Instructions : Use the legacy FTP service to access the pit archive.
[Composite] ImageSize : 900x600
[Composite] Megapixels : 0.540
SupraFast@JustTry:~$ |
```

Output yang diharapkan memuat Creator, Title, Description, dan Instructions. Bagian Description berisi username dan access phrase FTP yang menjadi clue tahap berikutnya.

- **Mengatur Ownership dan Permission**

Gambar hanya perlu dibaca oleh web server dan pemain. File tidak memerlukan execute permission.

```
SupraFast@JustTry:~$ sudo chown root:root \
/var/www/midnight/garage/supra-inspection.webp
sudo chmod 644 \
/var/www/midnight/garage/supra-inspection.webp
```

Verifikasi permission:

```
SupraFast@JustTry:~$ ls -l /var/www/midnight/garage/supra-inspection.webp
-rw-r--r-- 1 root root 62860 Sep  2 01:29 /var/www/midnight/garage/supra-inspection.webp
SupraFast@JustTry:~$ |
```

Hasil yang diawali dengan -rw-r--r-- dan tidak memiliki execute bit.

- **Menampilkan Gambar pada Halaman Garage**

Pastikan /var/www/midnight/garage/index.html mengarah ke ekstensi .webp, bukan .jpg. Contoh elemen HTML:

```
SupraFast@JustTry:~$ curl http://10.77.4.104/garage/
<!doctype html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>MK4 Inspection Archive</title>
</head>
<body>
  <h1>MK4 Pre-Race Inspection</h1>
  <p>Uploaded by the MK4 archive crew.</p>
  <p>Remember: vehicle media retains archival information even after publication.</p>
  supra-inspection.webp
</body>
</html>
SupraFast@JustTry:~$ |
```

Periksa seluruh referensi nama file:

```
SupraFast@JustTry:~$ grep -R 'supra-inspection' /var/www/midnight
/var/www/midnight/garage/index.html:      supra-inspection.webp
SupraFast@JustTry:~$ |
```

- **Verifikasi dari Server dan Attacker Peer**

Dari VPS, pastikan Nginx dapat menampilkan dan mengirim MIME type webp:

```
SupraFast@JustTry:~$ curl -I \
http://10.77.4.104/garage/supra-inspection.webp
HTTP/1.1 200 OK
Server: nginx/1.24.0 (Ubuntu)
Date: Wed, 02 Sep 2026 02:19:26 GMT
Content-Type: image/webp
Content-Length: 62860
Last-Modified: Wed, 02 Sep 2026 01:29:01 GMT
Connection: keep-alive
ETag: "6a977bdd-f58c"
Accept-Ranges: bytes
SupraFast@JustTry:~$ |
```

Respons yang diharapkan adalah HTTP/1.1 200 OK dengan Content-Type: image/webp.

Dari WSL atau Kali yang terhubung ke WireGuard, unduh gambar dan baca metadata:

```
~ Ilyas@HyperX mnt/..../ilyas >$ wget \
http://10.77.4.104/garage/supra-inspection.webp
file supra-inspection.webp
exiftool -a -G1 -s supra-inspection.webp

--2026-09-02 09:22:06-- http://10.77.4.104/garage/supra-inspection.webp
Connecting to 10.77.4.104:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 62860 (61K) [image/webp]
Saving to: 'supra-inspection.webp.1'

supra-inspection.webp.1      100%[=====] 61.39K  340KB/s  in 0.2s

2026-09-02 09:22:06 (340 KB/s) - 'supra-inspection.webp.1' saved [62860/62860]

supra-inspection.webp: RIFF (little-endian) data, Web/P image, XMP metadata, EXIF metadata, 900x600
[ExifTool]    ExifToolVersion    : 13.55
[System]      FileName           : supra-inspection.webp
[System]      Directory          : .
[System]      FileSize           : 63 kB
[System]      FileModifyDate      : 2026:09:01 07:39:16+07:00
[System]      FileAccessDate      : 2026:09:01 20:23:13+07:00
[System]      FileInodeChangeDate : 2026:09:01 20:23:13+07:00
[System]      FilePermissions     : -rwxrwxrwx
[File]        FileType           : Extended WEBP
[File]        FileTypeExtension   : webp
[File]        MIMEType            : image/webp
[File]        ExifByteOrder       : Big-endian (Motorola, MM)
[RIFF]        WebP_Flags          : XMP, EXIF
[RIFF]        ImageWidth          : 900
[RIFF]        ImageHeight         : 600
[RIFF]        VP8Version          : 0 (bicubic reconstruction, normal loop)
[RIFF]        ImageWidth          : 900
[RIFF]        HorizontalScale     : 0
[RIFF]        ImageHeight         : 600
[RIFF]        VerticalScale       : 0
[IF00]        XResolution         : 72
[IF00]        YResolution         : 72
[IF00]        ResolutionUnit      : inches
[IF00]        Artist              : SupraFast
[IF00]        YCbCrPositioning    : Centered
[XMP-x]       XMPToolkit          : Image::ExifTool 12.76
[XMP-dc]      Creator             : SupraFast
[XMP-dc]      Description         : Archive service: FTP | Crew: suprafast | Access phrase: ToyotaMK42JZ
[XMP-dc]      Title               : MK4 Pre-Race Inspection
[XMP-photoshop] Instructions     : Transfer the inspection record to the legacy pit archive.
[Composite]   ImageSize          : 900x600
[Composite]   Megapixels         : 0.540

~ Ilyas@HyperX mnt/..../ilyas >$
```

Pemain seharusnya menemukan service FTP, username suprafast, dan access phrase ToyotaMK42JZ. Informasi tersebut digunakan untuk melanjutkan ke tahap FTP.

9. Konfigurasi FTP

Struktur FTP:

/srv/midnight-ftp/pit-archive/

├── crew-handover.txt

└── vehicle-checklist.txt

Konfigurasi utama /etc/vsftdp.conf:

```
listen=YES
listen_ipv6=NO
listen_address=10.77.4.104
listen_port=21

anonymous_enable=NO
local_enable=YES
write_enable=NO

local_root=/srv/midnight-ftp
chroot_local_user=YES
allow_writeable_chroot=NO

pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40010
pasv_address=10.77.4.104

userlist_enable=YES
userlist_deny=NO
userlist_file=/etc/vsftdp.userlist

xferlog_enable=YES
use_localtime=YES
```

Daftar user yang diizinkan:

```
SupraFast@JustTry:~$ printf '%s\n' 'suprafast' | sudo tee /etc/vsftpd.userlist
suprafast
SupraFast@JustTry:~$ sudo chmod 644 /etc/vsftpd.userlist
SupraFast@JustTry:~$ |
```

Karena akun menggunakan /usr/sbin/nologin, shell tersebut ditambahkan ke /etc/shells agar autentikasi PAM FTP dapat berjalan:

```
SupraFast@JustTry:~$ grep -Fx '/usr/sbin/nologin' /etc/shells \
|| echo '/usr/sbin/nologin' | sudo tee -a /etc/shells
/usr/sbin/nologin
SupraFast@JustTry:~$ |
```

Permission data:

```
SupraFast@JustTry:~$ sudo chown -R root:root /srv/midnight-ftp
sudo find /srv/midnight-ftp -type d -exec chmod 755 {} \;
sudo find /srv/midnight-ftp -type f -exec chmod 644 {} \;
sudo systemctl enable --now vsftpd
sudo systemctl restart vsftpd
|
```

10. Konfigurasi SMB

Share disimpan pada /srv/samba/PitCrew, dan terdapat berkas didalam-Nya:

```
SupraFast@JustTry:~$ cd /srv/samba/PitCrew
SupraFast@JustTry:/srv/samba/PitCrew$ ls
r35-maintenance.txt  telemetry-client.conf  telemetry-readme.txt
SupraFast@JustTry:/srv/samba/PitCrew$ |
```

Akun samba dibuat:

```
SupraFast@JustTry:~$ sudo smbpasswd -a gazzoracing
sudo smbpasswd -e gazzoracing
|
```

Konfigurasi /etc/samba/smb.conf:

```
SupraFast@JustTry:~$ cat /etc/samba/smb.conf
[global]
    workgroup = WORKGROUP
    server role = standalone server
    security = user

    server min protocol = SMB2
    map to guest = never

    log file = /var/log/samba/log.%m
    max log size = 1000
    logging = file

    load printers = no
    printing = bsd
    printcap name = /dev/null
    disable spoolss = yes

[PitCrew]
    comment = Midnight Racing Pit Crew Share
    path = /srv/samba/PitCrew
    browseable = yes
    read only = yes
    guest ok = no
    valid users = gazzoracing
SupraFast@JustTry:~$ |
```

Test:

```
SupraFast@JustTry:~$ sudo testparm -s
sudo systemctl enable --now smbd
sudo systemctl restart smbd
Load smb config files from /etc/samba/smb.conf
Loaded services file OK.
Weak crypto is allowed by GnuTLS (e.g. NTLM as a compatibility fallback)

Server role: ROLE_STANDALONE

# Global parameters
[global]
    disable spoolss = Yes
    load printers = No
    log file = /var/log/samba/log.%m
    logging = file
    max log size = 1000
    printcap name = /dev/null
    security = USER
    server min protocol = SMB2
    server role = standalone server
    idmap config * : backend = tdb
    printing = bsd

[PitCrew]
    comment = Midnight Racing Pit Crew Share
    path = /srv/samba/PitCrew
    valid users = gazzoracing
Synchronizing state of smbd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable smbd
SupraFast@JustTry:~$
```

Samba listening pada semua local addresses, tetapi TCP/445 dibatasi oleh private lab network, firewall, dan Azure NSG.

11. Konfigurasi MariaDB

bind-address pada /etc/mysql/mariadb.conf.d/50-server.cnf diubah menjadi:

```
bind-address = 10.77.4.104
```

Service direstart:

```
SupraFast@JustTry:~$ sudo systemctl enable --now mariadb
sudo systemctl restart mariadb
Synchronizing state of mariadb.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mariadb
SupraFast@JustTry:~$
```

Database dibuat:

```
MariaDB [(none)]> CREATE DATABASE IF NOT EXISTS race_telemetry;
Query OK, 0 rows affected, 1 warning (0.000 sec)

MariaDB [(none)]> USE race_telemetry;
rds (
  id INT PRIMARY KEY AUTO_INCREMENT,
  vehicle VARCHAR(100) NOT NULL,
  lap_time VARCHAR(20) NOT NULL
)
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [race_telemetry]>
MariaDB [race_telemetry]> CREATE TABLE IF NOT EXISTS approved_drivers (
  id INT PRIMARY KEY AUTO_INCREMENT,
  ssh_driver VARCHAR(50) NOT NULL,
  vehicle VARCHAR(100) NOT NULL,
  access_phrase VARCHAR(100) NOT NULL,
  status VARCHAR(50) NOT NULL
);
Query OK, 0 rows affected, 1 warning (0.000 sec)

MariaDB [race_telemetry]>
MariaDB [race_telemetry]> CREATE TABLE IF NOT EXISTS lap_records (
  id INT PRIMARY KEY AUTO_INCREMENT,
  vehicle VARCHAR(100) NOT NULL,
  lap_time VARCHAR(20) NOT NULL,
  track VARCHAR(100) NOT NULL
);
Query OK, 0 rows affected, 1 warning (0.000 sec)

MariaDB [race_telemetry]> SHOW WARNINGS;
+-----+-----+-----+
| Level | Code | Message                                     |
+-----+-----+-----+
| Note  | 1050 | Table 'lap_records' already exists |
+-----+-----+-----+
1 row in set (0.000 sec)

MariaDB [race_telemetry]> |
```

Akun telemetry_reader hanya diberi hak baca:

```
MariaDB [race_telemetry]> GRANT SELECT ON race_telemetry.*
→ TO 'telemetry_reader'@'10.77.%';
Query OK, 0 rows affected (0.002 sec)

MariaDB [race_telemetry]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [race_telemetry]> |
```

Hak diverifikasi menggunakan:

```
MariaDB [race_telemetry]> SHOW GRANTS FOR 'telemetry_reader'@'10.77.%';
+-----+-----+
| Grants for telemetry_reader@10.77.% |
+-----+-----+
| GRANT USAGE ON *.* TO 'telemetry_reader'@'10.77.%' IDENTIFIED BY PASSWORD '*7906E9083B586F7C94FB03957BC54742D8A07103' |
| GRANT SELECT ON 'race_telemetry'.* TO 'telemetry_reader'@'10.77.%' |
+-----+-----+
2 rows in set (0.000 sec)

MariaDB [race_telemetry]> |
```

12. Konfigurasi SSH Foothold

Konfigurasi tambahan dibuat pada:

```
SupraFast@JustTry:~$ cat /etc/ssh/sshd_config.d/90-midnight-lab.conf
PasswordAuthentication yes
PermitRootLogin no
SupraFast@JustTry:~$ |
```

Dan clue terakhir ditempatkan pada:

/home/gtracing/final-inspection.txt

```
gtracing@JustTry:~$ cat /home/gtracing/final-inspection.txt
```

Clue didalam-Nya tidak menyebut nama atau lokasi SUID secara langsung. Pemain diminta memeriksa unusual locally installed utilities dan Linux permissions.

13. Final Flag dan Controlled SUID Reader

Flag disimpan pada /root/final-race-report.txt

```
root@JustTry:/home/SupraFast# cat /root/final-race-report.txt
MIDNIGHT RACING FINAL REPORT

Vehicle : R35 Track Car
Status  : Cleared for the final lap

MBC{Midnight_Pitlane_Final_Lap}
root@JustTry:/home/SupraFast# |
```

Permission:

```
SupraFast@JustTry:~$ sudo chown root:root /root/final-race-report.txt
sudo chmod 600 /root/final-race-report.txt
```

Controlled reader ditempatkan pada:

```
SupraFast@JustTry:~$ /opt/midnight-racing/bin/race-report-reader
```

Ownership dan SUID:

```
SupraFast@JustTry:~$ sudo chown root:root /opt/midnight-racing/bin/race-report-reader
sudo chmod 4755 /opt/midnight-racing/bin/race-report-reader
```

Program hanya membuka fixed path /root/final-race-report.txt dalam mode read-only. Program tidak menerima arbitrary path, tidak menjalankan shell, dan tidak mengubah sistem.

Cek:

```
root@JustTry:/home/SupraFast# sudo -u gtracing cat /root/final-race-report.txt
sudo -u gtracing /opt/midnight-racing/bin/race-report-reader
sudo -u gtracing test -w /opt/midnight-racing/bin/race-report-reader \
    && echo FAIL || echo PASS
cat: /root/final-race-report.txt: Permission denied
MIDNIGHT RACING FINAL REPORT

Vehicle : R35 Track Car
Status  : Cleared for the final lap

MBC{Midnight_Pitlane_Final_Lap}
PASS
root@JustTry:/home/SupraFast#
```

14. Firewall

Sebelum mengaktifkan UFW, akses SSH administrator harus diizinkan terlebih dahulu.

```
SupraFast@JustTry:~$ sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow in on eth0 from ADMIN_PUBLIC_IP to any port 22 proto tcp
sudo ufw allow in on mbc-las-w4 to 10.77.4.104 port 21 proto tcp
sudo ufw allow in on mbc-las-w4 to 10.77.4.104 port 22 proto tcp
sudo ufw allow in on mbc-las-w4 to 10.77.4.104 port 80 proto tcp
sudo ufw allow in on mbc-las-w4 to 10.77.4.104 port 445 proto tcp
sudo ufw allow in on mbc-las-w4 to 10.77.4.104 port 3306 proto tcp
sudo ufw allow in on mbc-las-w4 to 10.77.4.104 port 40000:40010 proto tcp
```

Setelah rule terapkan:

```
SupraFast@JustTry:~$ sudo ufw show added
sudo ufw enable
sudo ufw status numbered
```

Sesi SSH admin yang sedang aktif tidak ditutup sampai akses dari terminal kedua berhasil diuji.

15. Verifikasi Akhir

[illegible]

Hasil:

- Semua service aktif.
- Lima port challenge dapat diakses melalui WireGuard.
- gtracing tidak memiliki sudo rule khusus.
- Direktori dan binary challenge tidak writable oleh gtracing.
- Hanya custom SUID reader yang menjadi intended privilege path.
- Final flag hanya berada di final location.

16. Kesimpulan

VPS Azure berhasil digunakan sebagai target private CTF melalui WireGuard. Challenge menyediakan lima service yang mendukung clue trail, foothold Linux non-root, dan satu controlled SUID privilege-escalation proof.